Dayeon Yoo

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Education

University of New Mexico, M.S. in Computer Engineering — Aug 2025 – May 2027
Quantum Information Science emphasis; GPA: 4.06/4.33
Advisor: Prof. Milad Marvian

University of New Mexico, Exchange Student in Computer Science; GPA: 4.2/4.33 — Aug 2024 – May 2025
Supported by the Korea-U.S. Student Exchange Program in STEM Fellowship

Dongguk University, B.S. in Multimedia Software Engineering; GPA: 4.33/4.5 — Mar 2020 – Aug 2025
Graduated with Highest Honors (ranked 1st of 40 students)

Research Experience

Research Assistant, Center for Quantum Information and Control, University of New Mexico — Aug 2025 – Present

Manuscript in preparation: "Accelerating Quantum LDPC Code Optimization Through Adversarial Games" (with Prof. Milad Marvian).

- Developed an adversarial-game framework for quantum low-density parity check (qLDPC) code optimization, in which a code optimizer iteratively updates the code against an adversarial environment that identifies minimum-weight undetectable errors.
- Optimized several hypergraph-product codes, improving $[[1600, 64, 6]]$ to $[[1600, 64, 12]]$ and $[[2500, 100, 10]]$ to $[[2500, 100, 14]]$ in under a minute; for $[[625, 25]]$, achieved distance $d = 9$ in 3.7 seconds, while a logical-error-rate-guided random-walk baseline reached only $d = 8$ after 49 hours.
- Applied the adversarial framework to bivariate bicycle codes with weight-6 and weight-8 stabilizer checks, exploring block lengths up to $n = 800$ and $n = 300$, respectively; discovered promising weight-6 codes including $[[112, 12, 8]]$ and $[[340, 16, \leq 18]]$, and a weight-8 code $[[84, 18, 7]]$.

QU-REACH Research Intern, University of New Mexico — May 2025 – Aug 2025

- Developed graph-search methods for optimizing sparse structures in hypergraph-product quantum LDPC codes and benchmarked candidate-generation strategies against random-walk and simulated-annealing baselines.

Data Science Research Assistant, UNM Department of Marketing — Oct 2024 – May 2025

- Built a multimodal ML pipeline for 200K marketing advertisements, extracting image, video, audio, and text features using OpenCV, YOLOv8, Hugging Face Transformers, and speech-processing models for downstream analysis of creative performance and audience engagement.
- Conducted statistical analysis of multimodal advertisement features to investigate their relationships with audience engagement and brand perception.

Undergraduate Researcher, University of New Mexico — Aug 2024 – May 2025

- Implemented and evaluated an unsupervised anomaly-detection pipeline for MeV ultrafast electron diffraction images using a convolutional autoencoder.
- Evaluated reconstruction-error-based anomaly detection on over 1,500 experimental diffraction images to support automated data-quality screening.

Research Manuscripts

D. Yoo, M. Marvian, "Accelerating Quantum LDPC Code Optimization Through Adversarial Games," *manuscript in preparation*.

Presentations and Posters

Accelerating Quantum LDPC Code Optimization Through Adversarial Games — 2026
Poster presented at SQuInT 2026

Property Testing — 2026
Presentation, CQuIC Summer Course: Quantum Learning Theory

Fast and Accurate Neural-Network Decoders for Quantum Error-Correcting Codes — 2026
Presentation, UNM QEC Course

Barium-137 Trapped-Ion Quantum Computing — 2026
Presentation, UNM Platforms of Quantum Tech Course

Optimizing Hypergraph Product Codes 2025
Poster presented at QU-REACH Poster Session

Technical Skills

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, C, HTML/CSS, MATLAB

QEC: qLDPC codes (HGP, BB), BP+OSD, Stim, Qiskit, Monte Carlo simulation

ML & Data: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, pandas, NumPy, SciPy, OpenCV

Tools & Frameworks: HPC, Docker, PostgreSQL, HDF5, Slurm, Linux, Git, AWS, GCP, React Native, Next.js, NestJS

Work Experience

Co-Founder & Software Engineer, SnapIT! Jun 2024 – May 2025

- Co-founded and launched an AI-powered four-cut photo booth app featuring AI background removal, image generation, and customizable photo frames; reached 2K+ App Store downloads and was selected as a Top 10 finalist in the Epson Innovation Challenge.
- Led full-stack development and Epson printer integration; secured B2B partnerships to deliver customized photo booth experiences at four local events in Albuquerque, managing live deployments and stakeholder feedback.

Software Engineer, Divers Nov 2022 – Feb 2024

- Led full-stack development of a cross-platform mobile app using React Native, Next.js, NestJS, PostgreSQL, and AWS, scaling the platform to 15K+ users.
- Diagnosed production crashes and API/network failures using application telemetry, optimized runtime and memory performance, and validated releases through automated, unit, and integration testing.

Selected Projects

NASA Space Apps Challenge – Fire Fighter (Website) *1st Place, 2024*

- Developed an interactive wildfire analytics platform using NASA Earth-observation APIs, implementing dynamic geospatial visualizations of wildfire activity and effects on air quality, biodiversity, and agriculture.

Microsoft GenAI Hackathon – RAG Q&A Bot (Website) *3rd Place, 2024*

- Developed a retrieval-augmented generation (RAG) system for government documents, combining document retrieval and generative AI to produce context-grounded responses.

Lobo Hackathon – LLM Legal Information Platform (Website) *Prize Winner, 2025*

- Developed an LLM-powered platform that processes U.S. government documents and generates accessible summaries to help nonprofit and public users understand complex legal information.

Naechinso – Friend-Endorsed Dating App (Website) *2022*

- Developed a dating app requiring a recommendation from a real-life friend for registration; presented the product at the 11th Central MakeUs Challenge Demo Day.

Honors, Awards & Fellowships

Korea-U.S. Student Exchange Program in STEM Fellowship 2024

Academic Merit Scholarship, Dongguk University 2021 – 2022
Awarded for ranking 1st in the major by GPA; received for three semesters.

Teaching Experience

Algorithm Study Leader, Hack The World, University of New Mexico 2024 – 2025

- Led weekly algorithm study sessions, teaching algorithmic concepts and facilitating peer presentations and discussions of problem-solving approaches.

Teaching Assistant, Java Programming, Dongguk University Mar 2022 – Jun 2022

- Delivered in-person Java programming instruction through live coding demonstrations and hands-on exercises.

Academic Tutor, Java Programming, Dongguk University Sep 2021 – Feb 2022

- Tutored students in Java programming, covering object-oriented programming, data structures, algorithms, and software design principles.

Leadership & Extracurricular Activities

Member, IEEE UNM Chapter – Staffed the IEEE booth at CES 2025 and CES 2026 2024 – Present

Member, Society of Women Engineers (SWE) UNM chapter – Attended WE24 and WE25 2024 – Present

Founder & President, Hack The World Software Engineering Club – University of New Mexico 2024 – 2025

Officer, Korean Language and Culture Club – University of New Mexico 2024 – 2025

Web Development Lead, Central MakeUs Challenge (CMC) Nov 2022 – Mar 2023

Led web development study groups and organized a 24-hour hackathon and Demo Day for the 12th cohort.